DIRECTORATE OF TECHNICAL EDUCATION, GUJARAT

GTU Diploma Engineering – Information Technology, Semester VI

Cloud & Data Center Technologies

Course Code: 4361602

Lab Manual – Sections J & K

Procedure, Output & Quiz (All 12 Practicals)

Compiled from the official GTU Lab Manual

Created By: Jatin Harijan

0

1 Practical 1 – Cloud Platform Architectures 2

2 Practical 2 – Cloud Organization & Role-Based Access Control 3

3 Practical 3 – Install VirtualBox / VMware 4

4 Practical 4 – Install Python in a Virtual Machine 5

5 Practical 5 – Type 2 Virtualization in VMware 7

6 Practical 6 – Desktop Virtualization with Chrome Remote Desktop 9

7 Practical 7 – Virtual SDN Lab Using Mininet 11

8 Practical 8 – Cloud Databases Comparison 13

9 Practical 9 – Cloud Storage Comparison 15

10 Practical 10 – Resource Management Simulation with CloudSim 17

11 Practical 11 – Secure File Sharing with CloudSim 19

12 Practical 12 – Creating and Executing a Docker Container 21

Quick Answer Reference 24

1

Objective: Sketch and analyse the architecture of OpenStack / Eucalyptus / OpenNebula / KVM.

Section J – Procedure & Output

OpenStack – Key Components:

- **Nova** – Compute: provisions and manages virtual machines.
- **Neutron** – Networking: manages virtual networks between VMs.
- **Glance** – Image Service: stores and retrieves VM disk images.
- **Cinder** – Block Storage: provides persistent storage volumes.
- **Swift** – Object Storage: scalable blob/object store (like S3).
- **Keystone** – Identity: handles authentication and RBAC for all services.
- **Horizon** – Dashboard: web GUI to manage all components.

Eucalyptus – Key Entities:

- **CLC** (Cloud Controller) – manages users and provides EC2/S3-compatible API.
- **Walrus** – S3-compatible object storage for VM images.
- **CC** (Cluster Controller) – manages a cluster of physical nodes.
- **NC** (Node Controller) – runs on each physical host; controls VM lifecycle.

KVM: A Linux kernel module that turns Linux itself into a Type-1 hypervisor. Each VM runs as a regular Linux process via QEMU. Requires Intel VT-x or AMD-V CPU support.

Expected Output: Draw a labelled block diagram in your answer book showing components and their connections (e.g. Keystone at centre connecting to Nova, Glance, Neutron, Cinder, Swift, and Horizon with labelled arrows).

Q1

OpenStack controls a data center's large pool of _____.

(a) Computing (b) Storage (c) Networking **(d) All of the above ✓**

Answer

(d) All of the above – OpenStack manages computing (Nova), storage (Swift/Cinder), and networking (Neutron) resources together.

Q2

How many types of cloud computing service models are there?

(a) 2 **(b) 3 ✓** (c) 4 (d) 5

Answer

(b) 3 – The three models are **IaaS** (Infrastructure), **PaaS** (Platform), and **SaaS** (Software) as a Service.

2

Objective: Create a Cloud Organization in Google Cloud / AWS with RBAC.

Section J – Procedure & Output

Steps to assign Admin role in Google Cloud:

1. Sign in to `admin.google.com` with a Super Admin account.
2. Go to **Menu** → **Directory** → **Users** and select the target user.
3. Click **Admin roles and privileges** → toggle *Super Admin* → **Save**.

Steps to create a Custom Role:

1. Go to **Menu** → **Account** → **Admin Roles** → **Create new role**.
2. Enter name and description → check required privileges → **Create Role**.
3. Click **Assign Role** → select user → choose organisational unit → **Done**.

RBAC Concept: Permissions are granted based on *roles*, not individuals.

- **Primitive roles:** Owner, Editor, Viewer (broad)
- **Predefined roles:** e.g. `roles/compute.admin` (service-specific)
- **Custom roles:** Defined by admin for specific needs

Expected Output: Screenshot of Admin Console showing the new custom role and its assigned user under **Admin Roles**.

Q1

True or False: Google manages every aspect of GCP customers' security?

(a) True **(b) False** ✓

Answer

(b) False – Google uses a **Shared Responsibility Model**. Google secures the infrastructure; customers are responsible for IAM, data security, and access configuration.

Q2

Services and APIs are enabled on a per _____ basis.

(a) Billing account (b) Organization (c) Folder **(d) Projects** ✓

Answer

(d) Projects – In Google Cloud, services and APIs are enabled individually per **Project**, which is the fundamental unit of resource and billing management.

3

Objective: Install VirtualBox/VMware and set up Linux or Windows as a guest OS.

Section J – Procedure & Output

Steps to Install VirtualBox and Create a VM:

1. Download from `virtualbox.org/wiki/Downloads` for your host OS.
2. Run the installer → click **Next** through all steps → accept network interface prompt → **Install**.
3. Open VirtualBox → click **New** → enter VM name, Type: Linux, Version: Ubuntu 64-bit.
4. Allocate RAM (minimum 1024 MB) → create VDI hard disk (Dynamically Allocated, 8–20 GB).
5. Go to **Settings** → **Storage** → attach Ubuntu `.iso` file → click **Start**.
6. Follow Ubuntu installer: choose language, timezone, username/password → **Install**.
7. Remove ISO after installation and restart the VM.

Expected Output: VirtualBox Manager showing the VM in the left panel. VM boots into Ubuntu desktop. Take a screenshot of both the Manager window and the running guest OS.

Q1

What is a virtual machine (VM)?

(a) A physical computer **(b) A software-based emulation of a physical computer** ✓
(c) A mobile device (d) A server Operating System

Answer

(b) Software-based emulation of a physical computer – A VM is created by a hypervisor (e.g. VirtualBox) that abstracts the real hardware, allowing multiple OS to run on one machine.

Q2

In virtualization, what is the guest operating system?

(a) A physical computer **(b) A software-based emulation of a physical computer** ✓
(c) A mobile device (d) A server Operating System

Answer

(b) Software-based emulation – The **Guest OS** runs *inside* the VM (e.g. Ubuntu). The **Host OS** is what runs on real hardware (e.g. Windows). The hypervisor sits between them.

4

Objective: Install Python inside a VM and execute simple programs.

Section J – Procedure & Output

Steps to Install Python and Run Programs:

1. Boot the Ubuntu VM. Open Terminal (`Ctrl+Alt+T`).
2. Check if Python is installed: `python3 -version`
3. If not installed:

```
sudo apt-get update
sudo apt-get install python3 python3-pip -y
```

4. Create a simple program file:

```
# hello.py
print("Hello, World!")

# arithmetic.py
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Sum = ", a + b)
print("Product = ", a * b)
```

5. Run: `python3 hello.py` and `python3 arithmetic.py`
6. Install libraries: `pip3 install numpy pandas scipy`

Expected Output:

```
$ python3 hello.py
Hello, World!

$ python3 arithmetic.py
Enter first number: 10
Enter second number: 5
Sum = 15
Product = 50
```

Q1

Which Python libraries are used for data analysis and scientific computations?

(a) NumPy (b) Pandas (c) SciPy **(d) All of the above ✓**

Answer

(d) All of the above – **NumPy** handles arrays, **Pandas** handles DataFrames, and **SciPy** provides scientific algorithms. All are core data science libraries.

Q2

Which function checks if all characters in a string are decimal?

(a) `upper()` **(b) `isdecimal()` ✓** (c) `swapcase()` (d) `title()`

Answer

(b) `isdecimal()` – Returns `True` if all characters are digits 0–9. Example: `"123".isdecimal()` → `True`; `"12.3".isdecimal()` → `False`.

5

Objective: Perform VM operations in VMware – start/stop, memory/storage allocation, snapshots, cloning, and file transfer.

Section J – Procedure & Output

(a) Start / Stop VM: Select VM in VMware → click green **Power On** (`Ctrl+B`). To stop: **VM** → **Power** → **Shut Down Guest** (graceful) or force-stop button.
(b) Allocate Memory & Storage: Power off VM → **VM** → **Settings** → **Hardware**:

- Memory: drag slider to desired RAM (e.g. 2048 MB).
- Hard Disk: click **Expand** → enter new size in GB → OK.

(c) Take Snapshot: VM → **Snapshot** → **Take Snapshot** → enter name (e.g. "Before Updates") → **Take Snapshot**. Revert via **Snapshot Manager** → **Go To**.
(d) Clone VM: Power off → **VM** → **Manage** → **Clone** → choose:

- **Full Clone** – independent complete copy.
- **Linked Clone** – shares parent disk; saves disk space.

Enter name → **Finish**. New VM appears in library.
(e) Transfer Files:

- **Drag & Drop:** Enable in VM → Settings → Options → Guest Isolation.
- **Shared Folder:** Settings → Options → Shared Folders → Add host folder. Access at `/mnt/hgfs/` inside guest.
- **SCP:** `scp file.txt user@192.168.x.x:/home/user/`

Expected Output: Screenshots showing VM settings (RAM/disk), Snapshot Manager with created snapshot, and Clone Wizard completion screen.

Q1

What is VMware?

(a) A programming language **(b) A virtualization software company ✓**
(c) An open-source OS (d) A hardware manufacturer

Answer

(b) A virtualization software company – VMware makes products like Workstation (Type 2) and ESXi (Type 1) for creating and managing virtual machines.

Q2

Which type of virtualization is VMware Workstation?

(a) Type 1 **(b) Type 2 ✓** (c) Type 3 (d) Type 4

Answer

(b) Type 2 – VMware Workstation is a **hosted hypervisor** that runs on top of an existing OS (e.g. Windows). ESXi is a Type 1 (bare-metal) hypervisor.

Q3

Where does a Type 2 hypervisor run?

(a) Directly on hardware **(b) On top of a host OS ✓** (c) In the cloud (d) On a network server

Answer

(b) On top of a host OS – A Type 2 hypervisor runs as an application on the host OS. The host manages hardware; the hypervisor creates and manages guest VMs above it.

6

Objective: Set up and use Chrome Remote Desktop for desktop virtualization.

Section J – Procedure & Output

Steps to Set Up Host Machine:

1. Open Chrome → go to `remotedesktop.google.com/access`
2. Under *Set up remote access* → click **Download** to install extension.
3. Click **Turn On** → name the PC (e.g. "Lab-PC") → set a 6-digit PIN → **Start**.

Steps to Connect from Client Machine:

1. Open Chrome → go to same URL above.
2. Under *Remote Devices*, click the host PC name → enter PIN → **Connect**.
3. Host desktop appears inside the browser window.

Key Features Available:

- **File Transfer:** Sidebar → Send / Receive files.
- **Clipboard Sharing:** Copy on local, paste on remote (auto-enabled).
- **Multi-Monitor:** Switch screens via sidebar.
- **Disconnect:** Sidebar → Disconnect.

Expected Output: Screenshot of host desktop visible inside Chrome browser on client machine. Host listed as **Online** on the remote access page.

Q1

What is Chrome Remote Desktop used for?

(a) Web browsing **(b) Desktop virtualization ✓** (c) Game streaming (d) Social media

Answer

(b) Desktop virtualization – It allows remote access and control of another computer's desktop through the Chrome browser.

Q2

Which browser is required for Chrome Remote Desktop?

(a) Firefox (b) Safari (c) Internet Explorer **(d) Chrome ✓**

Answer

(d) Chrome – It requires the Chrome browser and the Chrome Remote Desktop extension installed on both host and client machines.

Q3

What is the first step to enable remote connections on the host machine?

(a) Set up VPN (b) Install software **(c) Enable remote connections in CRD ✓** (d) Update OS

Answer

(c) Enable remote connections in Chrome Remote Desktop – Open CRD → Turn On → set a PIN. This makes the host discoverable by the client.

Q4

What is essential for security in Chrome Remote Desktop?

(a) Weak PIN (b) Avoid updates **(c) Regularly update Chrome and CRD ✓** (d) Disable PIN

Answer

(c) Regularly update Chrome and the CRD extension – Updates patch security vulnerabilities. A strong PIN must always be kept enabled.

7

Objective: Set up a virtual SDN lab using VirtualBox and Mininet.

Section J – Procedure & Output

Steps to Set Up Mininet SDN Lab:

1. Create Ubuntu VM in VirtualBox (min. 2 GB RAM, 20 GB disk – see Practical 3).
2. Update the system:

```
sudo apt-get update && sudo apt-get upgrade -y
```

3. Install Mininet:

```
sudo apt-get install mininet -y
```

4. Test the installation with a simple topology:

```
sudo mn
mininet> pingall
mininet> exit
```

5. Create custom topologies:

```
sudo mn --topo single,3     # 1 switch, 3 hosts
sudo mn --topo linear,4    # 4 switches in a line
sudo mn --topo tree,depth=2,fanout=3
```

6. Useful Mininet CLI commands:

```
mininet> nodes        # list all nodes
mininet> net          # show links
mininet> h1 ping h2   # test connectivity
```

Expected Output:

```
mininet> pingall
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

Q1

What is the purpose of a virtual SDN lab?

(a) Gaming **(b) Educational and testing ✓** (c) Social networking (d) Video streaming

Answer

(b) Educational and testing – A virtual SDN lab simulates network topologies for learning and testing SDN concepts without needing physical hardware.

Q2

Which tool is used to create virtual SDN labs for educational purposes?

(a) Wireshark (b) GNS3 **(c) Mininet** ✓ (d) Docker

Answer

(c) Mininet – Mininet simulates complete networks (hosts, switches, controllers) on a single Linux machine. It is the standard tool for SDN learning.

Q3

Which command starts Mininet with a specific topology?

(a) `mininet start` (b) `mn -create` **(c)** `sudo mn -topo` ✓ (d) `start lab -topology`

Answer

(c) `sudo mn -topo` – Example: `sudo mn -topo single,3` creates 1 switch and 3 hosts. `sudo` is needed for root privileges.

8

Objective: Study and compare major cloud databases.

Section J – Procedure & Output

Prepare a written comparison based on: engine, type, key features, and best use case.

Service	Type / Engine	Key Feature & Use Case
Amazon RDS	SQL (MySQL, PostgreSQL, Aurora...)	Managed, multi-AZ, auto backup. General-purpose relational DB.
Google Cloud SQL	SQL (MySQL, PostgreSQL, SQL Svr)	GCP-integrated, auto backup. GCP-native relational apps.
Azure SQL DB	SQL (SQL Server)	Horizontal + vertical scaling, advanced security. Microsoft-stack apps.
IBM Db2 on Cloud	SQL (Db2)	Enterprise-grade, columnar storage. Legacy / analytics workloads.
Firebase RT DB	NoSQL (JSON)	Real-time sync, offline support. Mobile/web real-time apps.
MongoDB Atlas	NoSQL (BSON Document)	Horizontal sharding, flexible schema. Unstructured data apps.
Oracle Auto DB	SQL (Oracle)	Self-driving (auto-tune/patch/scale). Mission-critical OLTP.

Expected Output: Written comparison table as above. Optionally, create a free-tier account on any one service (e.g. MongoDB Atlas) and screenshot the dashboard showing instance name, region, engine, and status.

Q1

What is the primary purpose of Amazon RDS?

(a) Real-time data sync **(b) Managed relational database service** ✓
(c) NoSQL document storage (d) In-memory caching

Answer

(b) Managed relational database service – RDS automates backups, patching, scaling, and HA across multiple engines (MySQL, PostgreSQL, Oracle, etc.).

Q2

What type of databases does Azure SQL Database support?

(a) Only MySQL (b) Only PostgreSQL **(c) Relational / SQL Server compatible** ✓ (d) Only NoSQL

Answer

(c) Relational with SQL Server compatibility – Azure SQL is fully managed, SQL Server-compatible, supports horizontal/vertical scaling with advanced security.

Q3

What is a key feature of MongoDB Atlas?

(a) Auto vertical scaling (b) Managed relational DB (c) SQL Server support **(d) Horizontal scaling with sharding** ✓

Answer

(d) Horizontal scaling with sharding – Atlas distributes data across multiple servers (sharding) and uses flexible BSON documents instead of fixed relational schemas.

9

Objective: Study and compare cloud storage services: Amazon S3, Google Cloud Storage, Azure Blob Storage, and IBM Cloud Object Storage.

Section J – Procedure & Output

Service	Storage Tiers	Key Feature & Best For
Amazon S3	Standard, IA, Glacier, Int-Tiering	11-nines durability, versioning, lifecycle policies. Data lakes, backups.
Google Cloud Str	Standard, Near-line, Coldline, Archive	Integrates with BigQuery. GCP analytics and ML datasets.
Azure Blob Str	Hot, Cool, Archive	Cost-optimized tiering, integrates with Azure Data Lake. Large unstructured data.
IBM Cloud Obj St	Standard, Vault, Cold Vault	Cross-region resiliency, S3-compatible API. Hybrid enterprise storage.

Key Concepts:

- **Object Storage:** Flat structure, each item = data + metadata + unique ID. Accessed via HTTP API. Scales to petabytes.
- **File Storage:** Hierarchical folders. Accessed via NFS/SMB. Better for frequent random access.
- **Bucket:** Container for objects (like a root folder).
- **Storage Tiers:** Hot/Standard = frequent access (costly); Archive/Glacier = rare access (cheap, slow retrieval).

Expected Output: Written table as above. Optionally, create a free S3 bucket, upload a file, and screenshot the bucket dashboard showing object name and size.

Q1

What is object storage and how does it differ from traditional file storage?

Answer

Object Storage stores data as flat objects (data + metadata + unique ID), accessed via HTTP API, scales to petabytes. **File Storage** uses a hierarchical folder structure, accessed via NFS/SMB, better for frequent random access. Object storage is preferred for backups, media, and logs.

Q2

What does S3 stand for?

(a) Simple SQL Service (b) Straight Storage Service **(c) Simple Storage Service ✓**
(d) Simplified Serial Sequence

Answer

(c) Simple Storage Service – Amazon S3 stores data in *buckets*, offers 11-nines durability, and is accessed via REST APIs.

10

Objective: Simulate resource management using the CloudSim framework.

Section J – Procedure & Output

Setup:

1. Install Java JDK: `sudo apt install default-jdk`
2. Install Eclipse IDE (or IntelliJ IDEA).
3. Download CloudSim JAR from `cloudbus.org/cloudsim/`.
4. In Eclipse: **New Java Project** → **Build Path** → **Add External JARs** → add CloudSim JAR.

Key Simulation Code (CloudSimExample1.java):

```
CloudSim.init(1, Calendar.getInstance(), false);
Datacenter dc = createDatacenter("Datacenter_0");
DatacenterBroker broker = new DatacenterBroker("Broker");
Vm vm = new Vm(0, broker.getId(), 1000, 1,
               512, 1000, 10000, "Xen",
               new CloudletSchedulerTimeShared());
Cloudlet cl = new Cloudlet(0, 400000, 1, 300, 300,
               new UtilizationModelFull(), ...);
broker.submitVmList(vmlist);
broker.submitCloudletList(cloudletList);
CloudSim.startSimulation();
```

Key CloudSim Classes:

- **Datacenter** – models physical data center hardware.
- **Host** – a physical server with CPU, RAM, BW.
- **Vm** – virtual machine with MIPS, RAM, storage.
- **Cloudlet** – a task submitted to a VM.
- **DatacenterBroker** – manages VMs and cloudlets on behalf of users.

Expected Output:

```
========== OUTPUT ==========
Cloudlet ID   STATUS    DC ID   VM ID   Time   Start   Finish
    0         SUCCESS     2       0      0.4     0.1     0.5
CloudSimExample1 finished!
```

Q1

What is CloudSim?

(a) A cloud storage service **(b) An open-source framework for simulating cloud environments ✓**
(c) A VM management platform (d) A cloud deployment tool

Answer

(b) Open-source framework for simulating cloud environments – Written in Java by CLOUDS Lab. It models data centers, VMs, and tasks in a controlled environment before real deployment.

Q2

What kinds of entities can you model using CloudSim?

(a) Only VMs (b) Only data centers **(c) Both data centers and VMs ✓** (d) Only workloads

Answer

(c) Both data centers and VMs – CloudSim models Datacenters, Hosts, VMs, Cloudlets (tasks), and Brokers – giving a complete cloud environment simulation.

11

Objective: Simulate a secure file-sharing system using CloudSim with AES encryption.

Section J – Procedure & Output

Setup: Same as Practical 10 (Eclipse + CloudSim JARs).
Create `SecureCloudlet.java`: Extends `Cloudlet` and adds AES encrypt/decrypt:

```
public String encrypt(String content) {
  Cipher cipher = Cipher.getInstance("AES");
  cipher.init(Cipher.ENCRYPT_MODE, generateKey());
  byte[] enc = cipher.doFinal(content.getBytes());
  return Base64.getEncoder().encodeToString(enc);
}
public String decrypt(String content) {
  Cipher cipher = Cipher.getInstance("AES");
  cipher.init(Cipher.DECRYPT_MODE, generateKey());
  return new String(cipher.doFinal(
    Base64.getDecoder().decode(content)));
}
```

In `SecureFileSimulation.java`: Create 5 SecureCloudlets, call `encrypt()` on each, submit to broker. Full code is in lab manual pages 58–62.
Security Components Simulated:

- **Authentication** – only authorised brokers can submit cloudlets.
- **Encryption (AES)** – file content is encrypted before processing.
- **Access Control** – broker assigns cloudlets to specific VMs only.

Expected Output:

```
Cloudlet ID   STATUS    DC ID   VM ID   Time   Start   Finish
    1         SUCCESS     2       0      1.0     0.1     1.1
    2         SUCCESS     2       0      1.0     1.1     2.1
    ...
SecureFileSimulation finished!
```

Q1

What is CloudSim?

(a) A secure file-sharing platform **(b) A cloud simulation framework ✓**
(c) An encryption algorithm (d) A data storage solution

Answer

(b) A cloud simulation framework – CloudSim simulates cloud environments without running real software. It is used to test policies before actual deployment.

Q2

What is the essential component ensuring only authorised users can access the system?

(a) Datacenter (b) Encryption **(c) Authentication ✓** (d) VM Scheduler

Answer

(c) Authentication – Authentication is the gatekeeper that verifies identity before granting access. Encryption protects data; authentication controls *who* can enter.

12

Objective: Create and run a container using Docker.

Section J – Procedure & Output

Step 1 – Install Docker (Ubuntu):

```
sudo apt-get install docker.io -y
sudo systemctl start docker
docker --version    # verify
```

Or use online lab: `labs.play-with-docker.com` (no install needed).

Step 2 – Run a pre-built container:

```
docker run hello-world
```

Step 3 – Create a Dockerfile:

```
FROM alpine:latest
CMD ["echo", "Hello World!"]
```

Step 4 – Build and Run:

```
docker build -t my-first-container .
docker run my-first-container
# Output: Hello World!
```

Useful Docker Commands:

Command	Description
`docker ps`	List running containers
`docker ps -a`	List all containers (including stopped)
`docker images`	List all local images
`docker stop <id>`	Stop a running container
`docker rm <id>`	Remove a stopped container
`docker rmi <id>`	Remove an image
`docker exec -it <id> bash`	Enter a running container shell

Key Concepts: Image (template) → Container (running instance). Containers share host OS kernel (lightweight vs VMs which carry a full guest OS).

Expected Output:

```
$ docker run my-first-container
Hello World!

$ docker images
REPOSITORY            TAG       SIZE
my-first-container    latest    7.34MB
```

Q1

What is Docker primarily used for?

(a) Virtualization **(b) Containerization ✓** (c) Cloud Storage (d) Networking

Answer

(b) Containerization – Docker packages apps with all dependencies into lightweight, portable containers that share the host OS kernel (unlike VMs which need a full OS).

Q2

In Docker, what is a Dockerfile used for?

(a) To run containers **(b) To build images** ✓ (c) To create networks (d) To manage volumes

Answer

(b) To build images – A Dockerfile contains instructions (FROM, RUN, CMD) that Docker uses to build a container image. Flow: **Dockerfile** → **Image** → **Container**.

12

Complete Answer Key

Prac.	Q	Question Summary	Answer
1	1	OpenStack controls pools of ___	(d) All of the above
1	2	Types of cloud service models	(b) 3
2	1	Google manages ALL GCP security?	(b) False
2	2	APIs enabled per ___ basis	(d) Projects
3	1	What is a Virtual Machine?	(b) SW emulation
3	2	What is the Guest OS?	(b) SW emulation
4	1	Python libraries for data science	(d) All of the above
4	2	Checks all chars are decimal	(b) isdecimal()
5	1	What is VMware?	(b) Virtualization company
5	2	VMware Workstation is Type ___	(b) Type 2
5	3	Where does Type 2 hypervisor run?	(b) On host OS
6	1	Chrome Remote Desktop used for?	(b) Desktop virtualization
6	2	Required browser?	(d) Chrome
6	3	First step to enable host access?	(c) Enable in CRD
6	4	Essential for CRD security?	(c) Regular updates
7	1	Purpose of virtual SDN lab?	(b) Educational